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J. D. Foley
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Bresenham's Algorithm with Grey Scale

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The control parameter in Bresenham's algorithm, when interpreted as a measure of distance from a straight line boundary, gives a weighted average of the intensities on each side of the line. This can be used to produce an aesthetically pleasing visual effect with modern display devices.

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In a recent paper, Crow [2] presents a filtering technique which smooths rough edges due to the finite resolution of graphic display devices. As Crow points out, this is important in the generation of animated sequences, for otherwise ripples appear to run along the edges of objects as their position changes. The purpose of this note is to show that a minor addition to Bresenham's algorithm [1], which is widely used for the control of digital plotting and display equipment, can be used to achieve a similar effect. This approach is also useful with scan line based conversion algorithms, or scan conversion algorithms for polygons. (It does not, however, deal with the problem, also considered by Crow, of small objects which can flash on and off distractingly in an animated sequence; nor will it work at sharp corners.)

In its usual form, Bresenham's algorithm operates with a control variable d, the sign bit of which is used to determine whether to output a square or diagonal move to a digital plotter in order to draw the smoothest possible approximation to a straight line of the form $y = mx + c$ (see [3] for a full treatment of the case when $c \neq 0$). For simplicity, the discussion is restricted here to the first octant, i.e., the gradient is in the range $0 < m < 1$; other octants can be dealt with similarly.

The variable d is associated with the quantity $mx + c - y$, evaluated at the point $(i + 1, j + \frac{1}{2})$, where the pen is currently positioned at (i, j). If d is negative, the point $(i + 1, j + \frac{1}{2})$ is situated above the intended line, and a horizontal square move is the preferred choice. If d is positive, a diagonal move is called for. A revised version of the algorithm is sketched in Figure 1. The variable d has been increased by the constant quantity $w = 1 - m$ throughout. This results in extra arithmetic for the test, but d now operates in the range $0 \leq d < 1$, which is convenient for the purpose of this note.

Choosing the points for illumination on a display

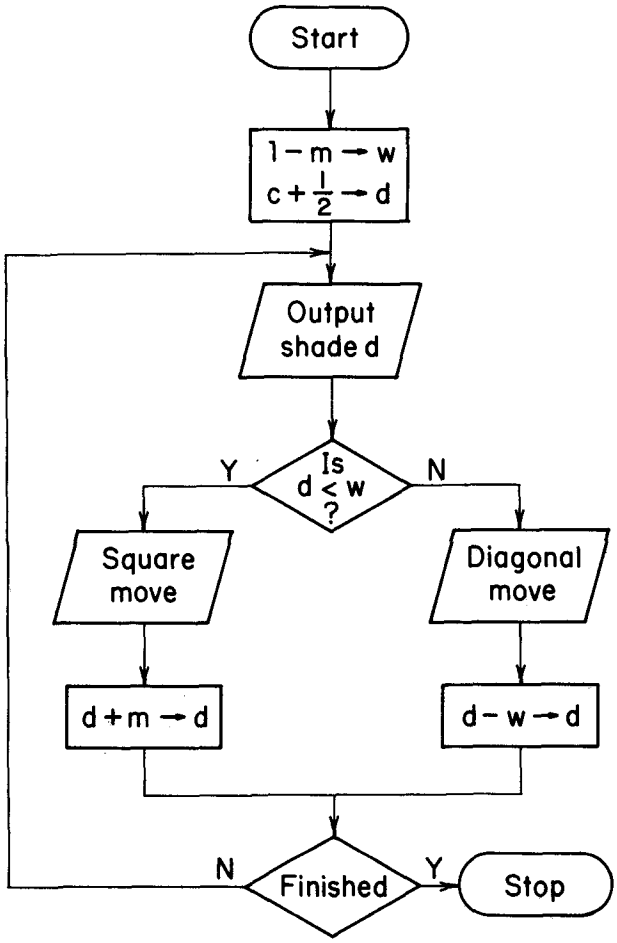


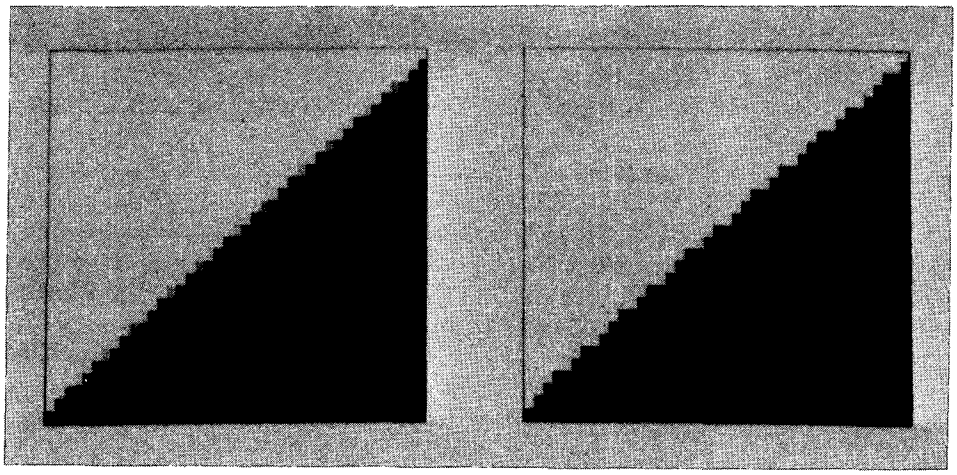
Fig. 1. A form of Bresenham's algorithm for the generation of the straight line $y = mx + c$, where $0 < m < 1$ (first octant), and $c < \frac{1}{2}$ (the origin is a sensible starting point). The algorithm has been modified so that d operates in the range $0 \leq d < 1$.

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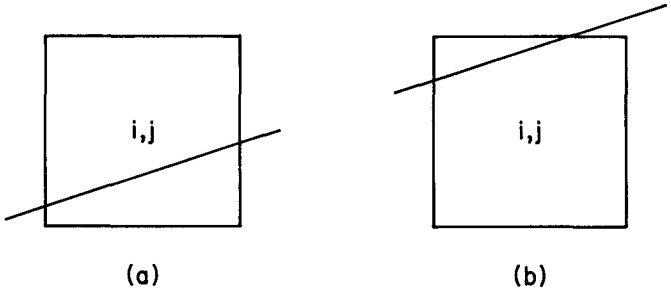
Fig. 2. The algorithm of Figure 1 has been used to trace out a straight line boundary from the origin at a gradient 5/7 between black above and white below. On the left, an intermediate grey is output at the meshpoints traced out by the algorithm. On the right, the extra information available in d is used to select an appropriate shade from black ($d = 0$) to white ($d = 1$) in order to mask the jagged edge by shading dark if the selected mesh point is above the true line, and by shading light if the selected mesh point is below the true line.



device is an analogous problem, and Bresenham's algorithm can be used to generate a straight line boundary between a dark and light area in grey scale or color work. In the example of Figure 2, left, the chosen points for a line of gradient $m = 5/7$ have been designated as an intermediate shade of grey.

On the right-hand side of Figure 2, the extra information available in d has been used to select a shade of grey varying from black for $d = 0$ to white for $d = 1$. The idea is to use the grey scale capability of modern graphic display devices to mask the excursions from the intended line caused by the point-to-point tracing of Bresenham's algorithm, shading dark if the mesh point is above the 5/7 line in Figure 2 and thus into the black region, and light if into the white region below the line. It turns out that this process selects exactly the same shade of grey as area integration when the intended boundary crosses a single pixel element, as shown in the case of Figure 3(a); the light area of the square below the line $y = mx + c$ is equal to $mi + c + \frac{1}{2} - j$, which is the same value as d from Figure 1 evaluated at $(i + 1, j + \frac{1}{2})$.

Fig. 3. Bresenham's error measure gives the same grey scale value as the fractional area of the pixel square (a) covered by the white side of the edge. In case (b), the error measure gives the appropriate shade of grey for smoothing the edge with a single pixel element, leaving the upper element completly black.



When the boundary involves two pixel elements, as in Figure 3(b), only the pixel element shown is shaded grey by the algorithm of Figure 1. The shade computed, however, is exactly appropriate for this treatment of the problem; the light area in the square shown is $1 - \{m(i - \frac{1}{2}) + c - j - \frac{1}{2}\}^2/(2m)$, and adding this to the triangular light area in the pixel element above, $\{m(i + \frac{1}{2}) + c - j - \frac{1}{2}\}^2/(2m)$, totals to $mi + c + \frac{1}{2} - j$ as before. Thus, the shading of the Figure 1 algorithm allows the same light transmission as pixel integration, without the computational overheads.

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